Project Synopsis

Project Title: SMART AGRICULTURE PLATFORM

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Enrollment No. / Roll No: 117

Course & Semester: MCA 1

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1. Introduction

Farming is a very important part of India's economy. But many farmers still use old ways for growing crops and managing their farms. The Smart Agriculture Platform helps farmers use modern technology to make better decisions. This system uses AI (Artificial Intelligence) and real-time data to give smart advice to farmers. It helps them choose the best crops, check weather reports, find plant diseases, and know market prices — all in one place. With this platform, farmers can save time, reduce losses, and earn more profit by farming in a smart and easy way.

2. Objectives

- To develop an AI-based web platform that helps farmers make data-driven agricultural decisions.

- To recommend the best crop based on soil nutrients, pH, temperature, humidity, and rainfall data.
- To provide weather forecasts and disease detection using machine learning models.
- To enable administrators to manage crop data, images, and information through an admin dashboard.
- To maintain a database for farmers' data, crop information, and recommendations.

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3. Problem Definition

Farmers face challenges such as unpredictable weather, poor soil management, low productivity, and lack of access to modern technology. Manual methods often lead to incorrect crop selection and economic losses. The **Smart Agriculture Platform** addresses these issues by offering intelligent tools that analyze environmental and soil parameters to suggest the most profitable crops and provide timely advisory support.

4. Scope of the Project

This platform can be used by farmers, agricultural experts, and government agencies to monitor and enhance agricultural productivity. It integrates AI-powered tools for crop recommendation, weather forecasting, and market price predictions. Future enhancements may include IoT-based soil sensors and mobile app integration for real-time notifications and remote monitoring.

5. System Requirements

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB

- Storage: Minimum 20 GB free space
- Internet Connection: Required for live updates

Software Requirements

- Operating System: Windows 10 or later
- Programming Language: Python, HTML, CSS, JavaScript
- Framework: Flask
- Database: MySQL
- Tools: Visual Studio Code / PyCharm
- Libraries: Flask-MySQLdb, OpenCV (for image processing), Scikit-learn, Pandas, NumPy

6. System Design / Methodology

The system follows these main modules:

1. **User Module:**
    - Farmers can register, log in, and input soil and weather data.
    - Receive crop recommendations and market insights.
2. **Admin Module:**
    - Add, edit, and manage crop data (name, image, duration, fertilizers, profit, and description).
    - Monitor user activity and data stored in the system.
3. **AI Crop Recommendation System:**
    - Processes user inputs (N, P, K, pH, temperature, humidity, rainfall).
    - Suggests the most suitable crop using an ML model.
4. **Database Management:**
    - Stores user details, weather data, crop information, and AI-generated recommendations.
5. **Frontend Interface:**
    - Responsive web pages for users and admin with easy navigation and real-time data updates.

7. Expected Outcome

- Farmers will receive accurate crop recommendations with visual details and profitability insights.
- Admins can manage and update crop information efficiently.
- Improved decision-making for farmers leading to better yield and sustainable farming practices.
- Centralized data storage for analysis and government policy development.

8. Conclusion

The **Smart Agriculture Platform** revolutionizes traditional farming by integrating artificial intelligence and data analytics. It helps farmers make informed choices regarding crop selection, soil management, and market trends. The system ensures efficiency, sustainability, and profitability in agriculture, contributing to smart and modern farming.

9. References

- https://python.org
- https://flask.palletsprojects.com
- https://www.mysql.com
- https://scikit-learn.org
- https://www.kaggle.com